

HYDROGENATED NBR: PATENT LANDSCAPE AND PROCESS SYNTHESIS REPORT

1. ABSTRACT

The provided patent set does not contain a direct, primary disclosure of hydrogenated nitrile butadiene rubber (HNBR) synthesis by conventional NBR hydrogenation chemistry. Instead, the evidence captures adjacent enabling technologies and downstream use contexts that are relevant to HNBR processing, modification, or application. EP2519350B1 is the most directly relevant because it explicitly mentions methods for making modified NBR and depolymerized HNBR using novel ruthenium metathesis catalysts, including a sequence in which NBR is first depolymerized at 30–100°C and then hydrogenated under high pressure at 60–150°C, or alternatively hydrogenation is performed first followed by catalyst addition at 60–150°C. This patent therefore provides the clearest process window in the manifest for hydrogenation-related rubber transformation, although it is not a dedicated HNBR preparation patent and the exact catalyst identity, pressure range, and conversion targets for HNBR are not fully resolved from the available evidence. The remaining patents are not HNBR synthesis patents but provide useful context: US11262338B2 focuses on modified thermoplastic resins with reduced oligomer content and improved Tg/Mz, VOC/FOG, and compatibility in rubber and adhesive formulations; US10294210B2 describes maleinated derivatives from functionalized monomers and olefin metathesis products, including hydrogenation and post-functionalization steps; EP2697186B1 discloses selective hydrogenation of terpene-derived feedstocks under staged hydrogenation conditions (e.g., 200°C and 50–200 kPa H₂ in later stages) to tune unsaturation; and EP3137587B1 is a detergent/biofilm patent with no meaningful HNBR synthesis relevance. Across the set, the dominant technical themes are controlled hydrogenation, staged or selective hydrogen delivery, catalyst-mediated transformation, and property tuning through unsaturation reduction or molecular-weight/oligomer control. For HNBR specifically, the evidence supports a process concept centered on NBR precursor transformation under hydrogenation conditions in the 60–150°C range, with catalyst selection and hydrogen pressure management being the key variables, but the manifest does not provide a complete, experimentally grounded HNBR recipe.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11262338B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example a: - Example Extracted Example Example 2.: - Example Extracted Example Example 2: - Example Extracted Example example 20: - Example Extracted Example Example 1.: - Example Extracted Example Example 2.: - Example Extracted Example Example 3.: - Example Extracted Example Example 4.: - Example Extracted Example Example 3: - Example Extracted Example Example 5.: - Example Extracted Example Example 6.: - Example Extracted Example Example 4: - Example Extracted Example Example 7.: - Example Extracted Example Example 7.: - Example Extracted Example Example 8.: - Example Extracted Example Example 9.: - Example Extracted Example Example 5.: - Example Extracted Example Example 10.: - Example Extracted Example Example 11.: - Example Extracted Example Example 10:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US10294210B2
- Patent Title: Maleinated derivatives
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4A: - Example Extracted Example Example 4B: - Example Extracted Example Example 4C: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9: - Example Extracted Example Example 10: - Example Extracted

Example Example 11: - Example Extracted Example Example 12: - Example Extracted Example
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Example Extracted Example Example 18:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: EP2697186B1
- Patent Title: Base oils and their methods of production
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 6.: - Example
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Example Example 29: - Example Extracted Example Example 48: - Example Extracted Example
Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: -
Example Extracted Example Example 4:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 3: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 4: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 5: - Example Extracted Example Example 2: - Example Extracted Example Example 6: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 7: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 8: - Example Extracted Example Example 1: - Example Extracted Example Example 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: EP3137587B1
- Patent Title: Detergent composition
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example 2: - Example Extracted Example example 3: - Example Extracted Example example 0: - Example Extracted Example example a: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 1.: - Example Extracted Example Example 3: - Example Extracted Example Example 1: - Example Extracted Example Example 1.: - Example Extracted Example Example 4:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2519350B1 is the only primary patent in the manifest that explicitly names HNBR; it discloses depolymerization of NBR followed by hydrogenation, or hydrogenation followed by catalyst addition, at 60–150°C, but does not provide a fully extracted monomer-content or

conversion dataset for HNBR.

- EP2697186B1 and EP2519350B1 both emphasize staged/selective hydrogenation concepts, but EP2697186B1 applies them to terpene feedstocks and reports explicit second-stage conditions such as 200°C with 50–200 kPa H₂, whereas EP2519350B1 applies them to rubber transformation and gives broader temperature windows.
- US10294210B2 and EP2519350B1 both involve catalyst-enabled transformations followed by hydrogenation, but US10294210B2 is focused on maleinated derivatives from olefinic monomers and uses peroxide/thermal ene chemistry rather than rubber hydrogenation; its examples show 150–200°C thermal processing and vacuum stripping rather than NBR-specific conditions.
- US11262338B2 is formulation-centric rather than HNBR-centric: it uses modified thermoplastic resins with reduced oligomer content to improve compatibility, VOC/FOG, and adhesive/rubber performance, whereas EP2519350B1 is a true rubber transformation patent aimed at NBR/HNBR modification.
- EP3137587B1 is not a synthesis patent for HNBR; it is a detergent composition patent and provides no relevant polymerization, hydrogenation, or NBR precursor data.
- Across the manifest, catalyst-mediated selectivity and post-reaction purification are recurring themes, but only EP2519350B1 directly links those themes to NBR/HNBR processing.
- The newer hydrogenation-oriented disclosures (EP2697186B1, EP2519350B1) emphasize staged control of hydrogen delivery and temperature, while the other patents are older or adjacent and focus on formulation/property optimization rather than direct HNBR synthesis.

4. CONCLUSION

The evidence base for Hydrogenated NBR in this manifest is limited but still technically informative. EP2519350B1 is the key patent because it explicitly addresses HNBR-related processing and indicates that NBR can be depolymerized and then hydrogenated, or hydrogenated first and then catalytically processed, within a 60–150°C window. Although the exact catalyst system, hydrogen pressure, and degree of hydrogenation are not fully extracted from the available text, the patent supports the general strategy of using a selective ruthenium-catalyzed process with staged operation to control backbone saturation and rubber properties. The broader patent set reinforces that successful hydrogenation-related polymer processing depends on tight control of reaction staging, catalyst choice, and downstream purification or fractionation. For practical HNBR synthesis based on the available evidence, the most defensible recommendation is a controlled hydrogenation workflow starting from NBR, using moderate-to-elevated temperature hydrogenation in the 60–150°C range, with pressure and catalyst loading selected to avoid over-processing while achieving the desired residual unsaturation. Because the manifest does not provide a complete HNBR recipe, any final process design should be validated experimentally against target residual double-bond content, molecular-weight retention, and elastomer performance.

5. REFERENCES

- US11262338B2 | Modified resins and uses thereof | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11262338B2/en>
- US10294210B2 | Maleinated derivatives | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10294210B2/en>
- EP2697186B1 | Base oils and their methods of production | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP2697186B1/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP3137587B1 | Detergent composition | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP3137587B1/en>